

Project: GrantServe - Research Grants & Academic Funding Management System

1. Introduction

GrantServe is a web-based platform designed for universities, research institutions, and government funding agencies to manage research grants, academic funding, and compliance. It enables researchers to apply for grants, administrators to oversee funding programs, reviewers to evaluate proposals, and compliance officers to monitor adherence to funding policies.

The system supports workflows for researcher registration, grant application submission, proposal review, funding disbursement, compliance monitoring, and analytics. It ensures transparency and accountability by maintaining audit trails, dashboards, and performance metrics across research funding operations.

Actors / Users

- **Researcher:** Registers, submits grant applications, tracks funding status.
- **Reviewer:** Evaluates proposals, provides feedback, recommends approval/rejection.
- **Program Manager:** Oversees grant programs, monitors budgets, tracks performance.
- **Administrator:** Configures workflows, manages users, oversees reporting.
- **Compliance Officer:** Ensures adherence to funding policies, audits records.
- **Government Auditor:** Reviews compliance reports, monitors grant utilization.

2. Module Overview

- 2.1 Identity & Access Management
- 2.2 Researcher Registration & Profile Management
- 2.3 Grant Application & Proposal Submission
- 2.4 Proposal Review & Evaluation
- 2.5 Funding Program & Budget Management
- 2.6 Disbursement & Payment Tracking
- 2.7 Compliance & Audit Management
- 2.8 Reporting & Analytics
- 2.9 Notifications & Alerts

3. Architecture Overview

- **Frontend:** Angular or React for responsive dashboards.

- **Backend:** REST API-based microservices (Identity, Researchers, Applications, Reviews, Programs, Disbursements, Compliance, Reports).
- **Database:** Relational DB (MySQL/PostgreSQL/SQL Server).
- **Deployment:** Cloud/on-prem with API gateway, WAF, centralized logging.

4. Module Wise Design

4.1 Identity & Access Management

Features:

Provides secure authentication, role-based access control, and audit logging.

Entities:

- User(UserID, Name, Role [Researcher/Reviewer/Manager/Admin/Compliance/Auditor], Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)

4.2 Researcher Registration & Profile Management

Features:

Registers researchers, maintains profiles, validates documents.

Entities:

- Researcher(ResearcherID, Name, DOB, Gender, Institution, Department, ContactInfo, Status)
- ResearcherDocument(DocumentID, ResearcherID, DocType [IDProof/Publication], FileURI, UploadedDate, VerificationStatus)

4.3 Grant Application & Proposal Submission

Features:

Researchers submit grant applications, track status, upload proposals.

Entities:

- GrantApplication(ApplicationID, ResearcherID, ProgramID, Title, SubmittedDate, Status)
- Proposal(ProposalID, ApplicationID, FileURI, SubmittedDate, Status)

4.4 Proposal Review & Evaluation

Features:

Reviewers evaluate proposals, provide feedback, recommend approval/rejection.

Entities:

- Review(ReviewID, ProposalID, ReviewerID, Score, Comments, Date, Status)
- Evaluation(EvaluationID, ApplicationID, Result [Approved/Rejected], Date, Notes)

4.5 Funding Program & Budget Management

Features:

Manages grant programs, tracks budgets, monitors performance.

Entities:

- Program(ProgramID, Title, Description, StartDate, EndDate, Budget, Status)
- Budget(BudgetID, ProgramID, AllocatedAmount, SpentAmount, RemainingAmount, Status)

4.6 Disbursement & Payment Tracking

Features:

Tracks grant disbursements, monitors payments, reconciles budgets.

Entities:

- Disbursement(DisbursementID, ApplicationID, Amount, Date, Status)
- Payment(PaymentID, DisbursementID, Method [Bank/Wallet], Date, Status)

4.7 Compliance & Audit Management

Features:

Ensures adherence to funding policies, audits records.

Entities:

- ComplianceRecord(ComplianceID, EntityID, Type [Application/Program], Result, Date, Notes)
- Audit(AuditID, OfficerID, Scope, Findings, Date, Status)

4.8 Reporting & Analytics

Features:

Generates dashboards and reports on applications, reviews, disbursements, and compliance.

Entities:

- Report(ReportID, Scope [Application/Program/Researcher], Metrics, GeneratedDate)

4.9 Notifications & Alerts

Features:

Sends alerts for application status, review updates, disbursement notifications, compliance deadlines.

Entities:

- Notification(NotificationID, UserID, EntityID, Message, Category [Application/Review/Disbursement/Compliance], Status, CreatedDate)

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB.
- **Production:** Cloud/on-prem deployment, API gateway, WAF, centralized logging.

6. Database Design

Tables:

- User(UserID, Name, Role, Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)
- Researcher(ResearcherID, Name, DOB, Gender, Institution, Department, ContactInfo, Status)
- ResearcherDocument(DocumentID, ResearcherID, DocType, FileURI, UploadedDate, VerificationStatus)
- GrantApplication(ApplicationID, ResearcherID, ProgramID, Title, SubmittedDate, Status)
- Proposal(ProposalID, ApplicationID, FileURI, SubmittedDate, Status)
- Review(ReviewID, ProposalID, ReviewerID, Score, Comments, Date, Status)
- Evaluation(EvaluationID, ApplicationID, Result, Date, Notes)
- Program(ProgramID, Title, Description, StartDate, EndDate, Budget, Status)
- Budget(BudgetID, ProgramID, AllocatedAmount, SpentAmount, RemainingAmount, Status)
- Disbursement(DisbursementID, ApplicationID, Amount, Date, Status)
- Payment(PaymentID, DisbursementID, Method, Date, Status)
- ComplianceRecord(ComplianceID, EntityID, Type, Result, Date, Notes)
- Audit(AuditID, OfficerID, Scope, Findings, Date, Status)
- Report(ReportID, Scope, Metrics, GeneratedDate)
- Notification(NotificationID, UserID, EntityID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Researcher Portal:** Grant application submission, proposal uploads, disbursement tracking.
- **Reviewer Dashboard:** Proposal evaluation, feedback submission, scoring.
- **Program Manager Console:** Program oversight, budget monitoring, reporting.
- **Administrator Panel:** Workflow setup, compliance monitoring, reporting.
- **Compliance Officer Console:** Policy monitoring, audits, compliance reporting.
- **Government Auditor Dashboard:** Grant utilization review, compliance monitoring.

8. Non-Functional Requirements

- **Performance:** Handle 50,000 concurrent users across research networks.
- **Security:** Role-based access, encrypted data storage, immutable audit logs.
- **Scalability:** Support multi-institution, multi-program rollout.
- **Availability:** 99.9% uptime.
- **Maintainability:** Modular microservices, API versioning, automated migrations.
- **Observability:** Centralized logging, KPIs (application approval rate, grant utilization, compliance adherence).

9. Assumptions & Constraints

- Initial rollout for a single funding program before multi-institution expansion.
- Notifications limited to in-app and SMS/email alerts.
- Fully implementable using Java Spring Boot/ASP.NET Core, Angular/React, and relational DB.